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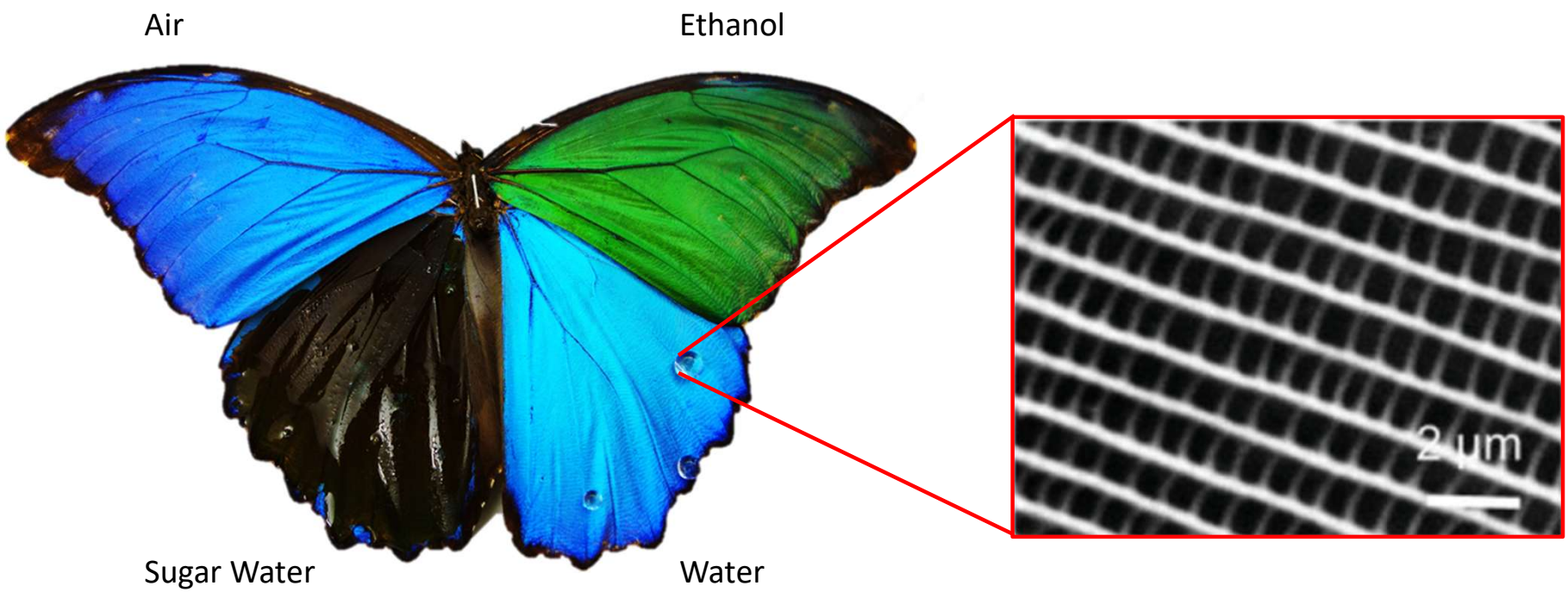
Negative Refraction Dielectric Metasurface for High Spatial Resolution Indirect X-Ray Detector

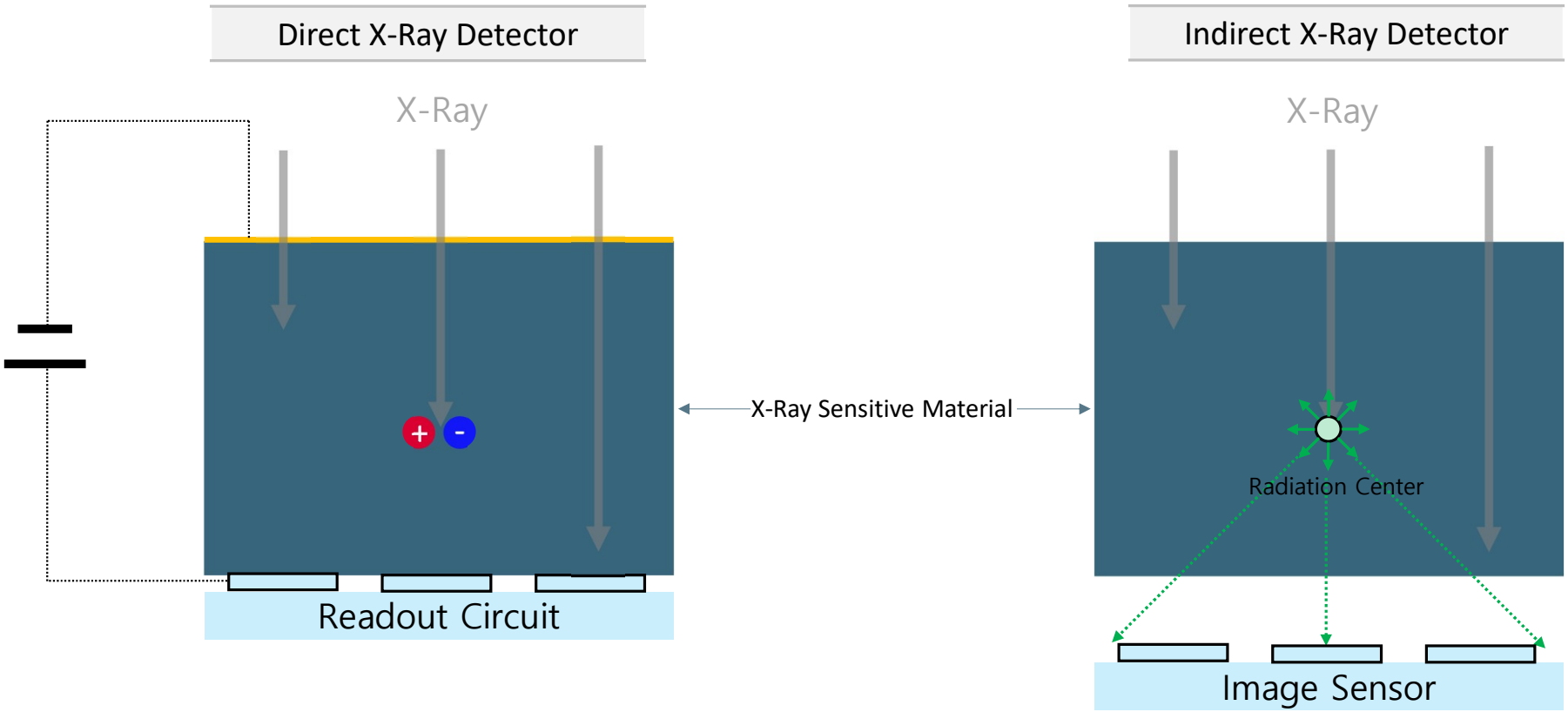
Last Updated: 202511

Introduction: Metasurface



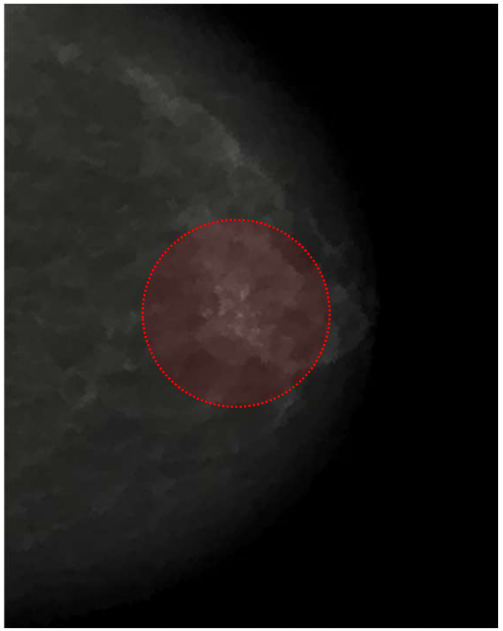
Bober, Brittany A., et al. "Investigating nanoscopic structures on a butterfly wing to explore solvation and coloration." *Journal of Chemical Education* 95.6 (2018): 1004-1011.
Cao, Xiaobao, et al. "Replicating the Cynandra opis butterfly's structural color for bioinspired bigrating color filters." *Advanced Materials* 34.9 (2022): 2109161.

Introduction: Classification for X-Ray Detector

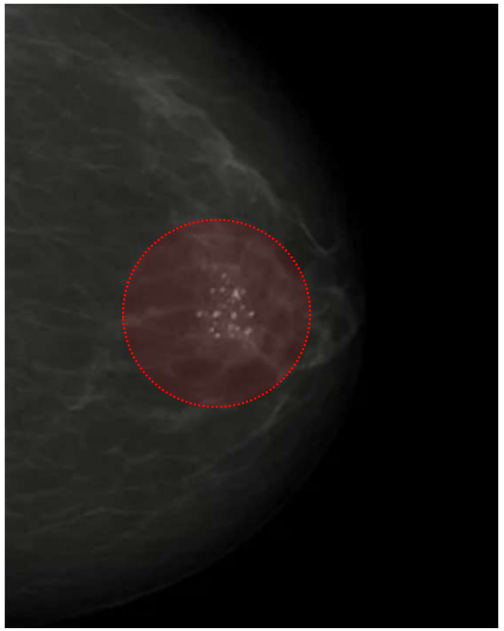


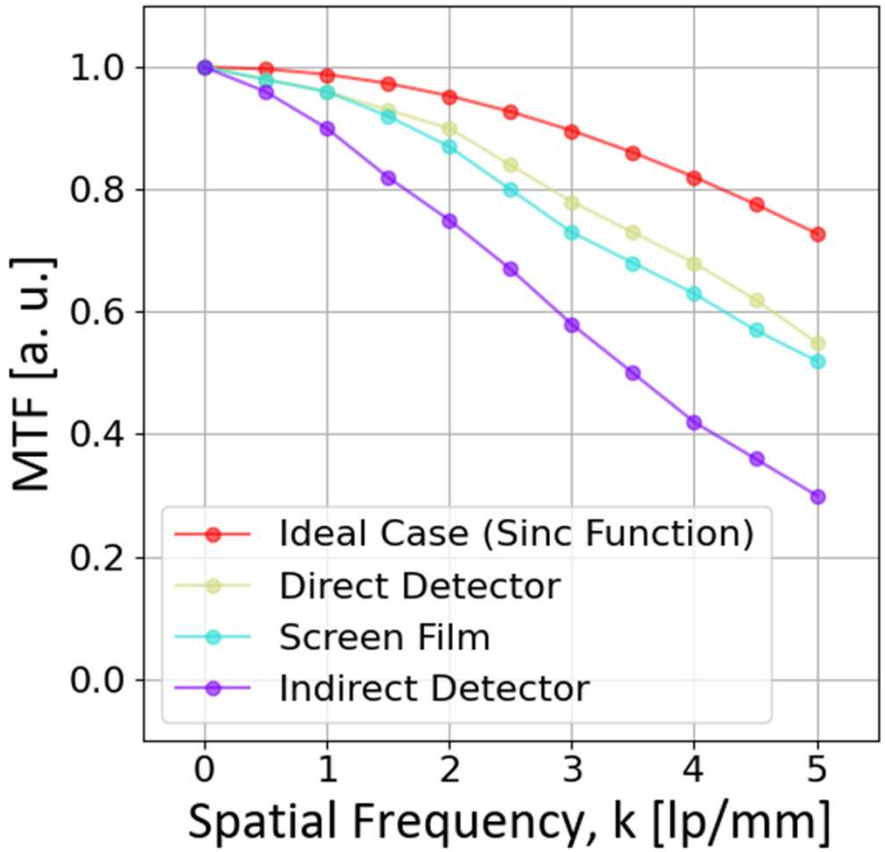
Introduction: Requirements for Higher Spatial Resolution

Low Resolution



High Resolution



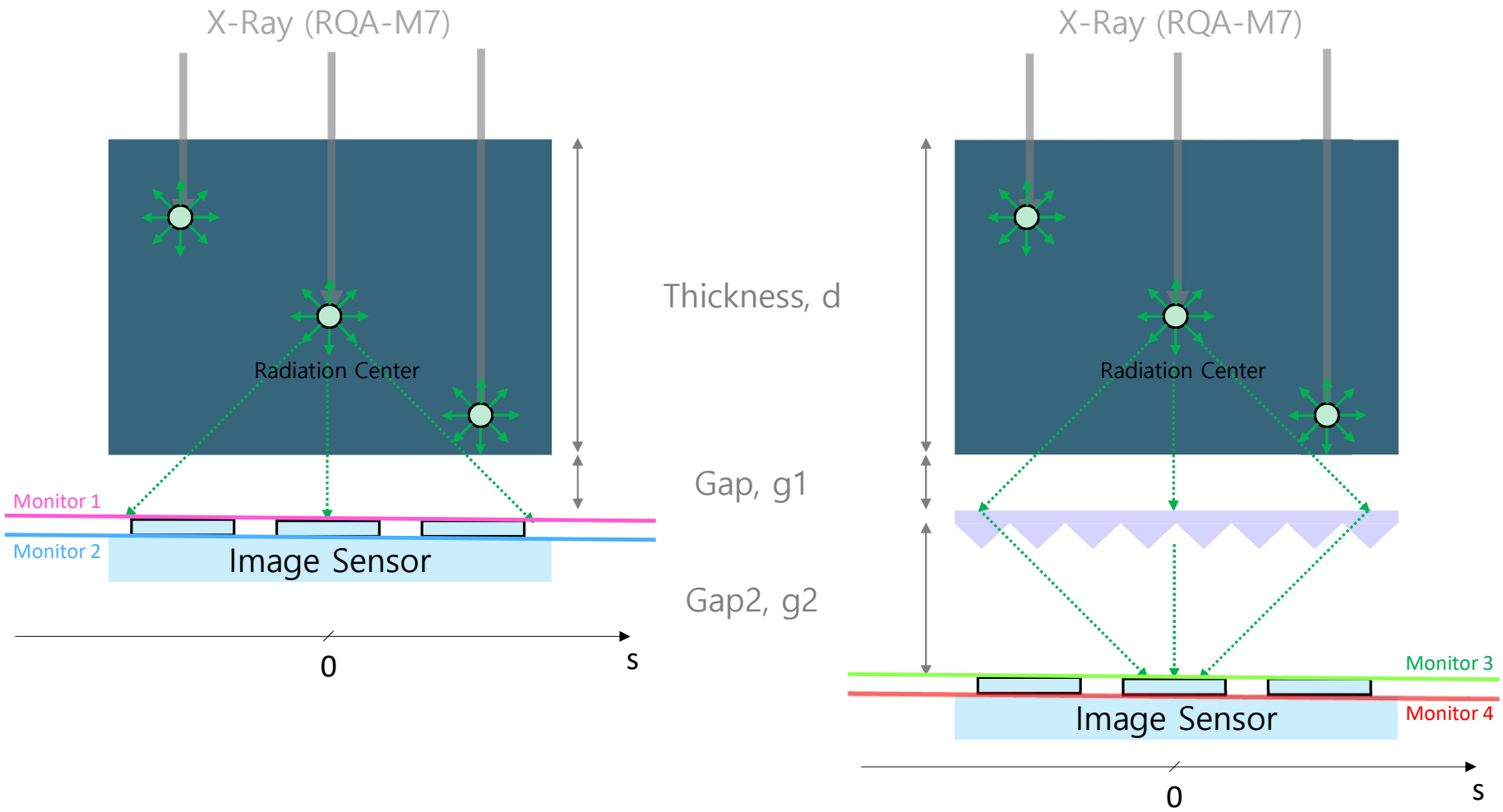


Microcalcification (Sign of Breast Cancer)

Why High-Brightness Monitors Matter in Mammography

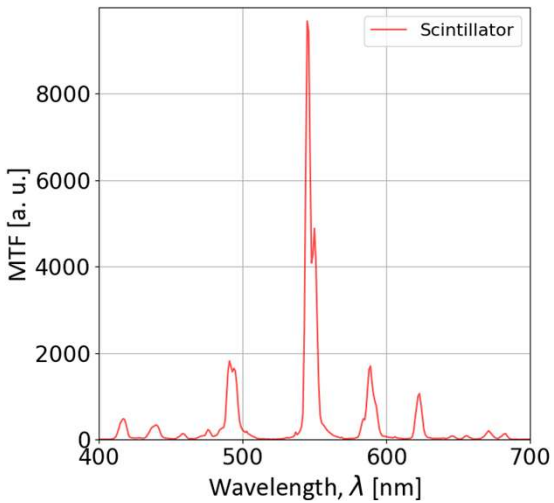
Yaffe, Martin J. "Detectors for digital mammography." *Digital mammography*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2010. 13-31.

Simulation Scheme

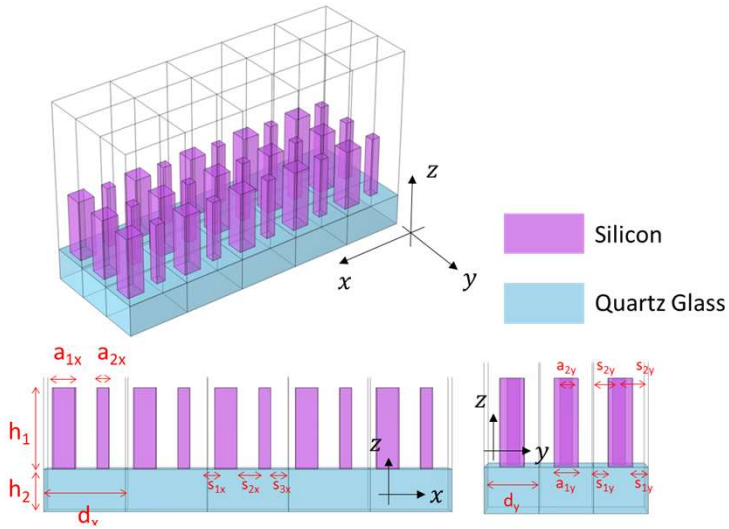


Metasurface Design

Scintillator Spectrum

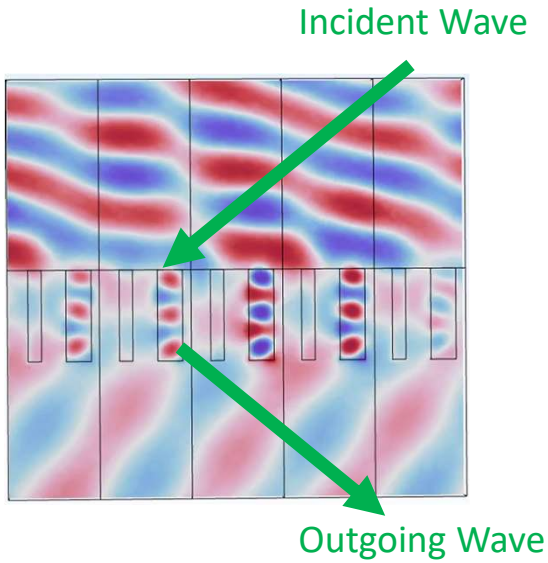


Design Parameters



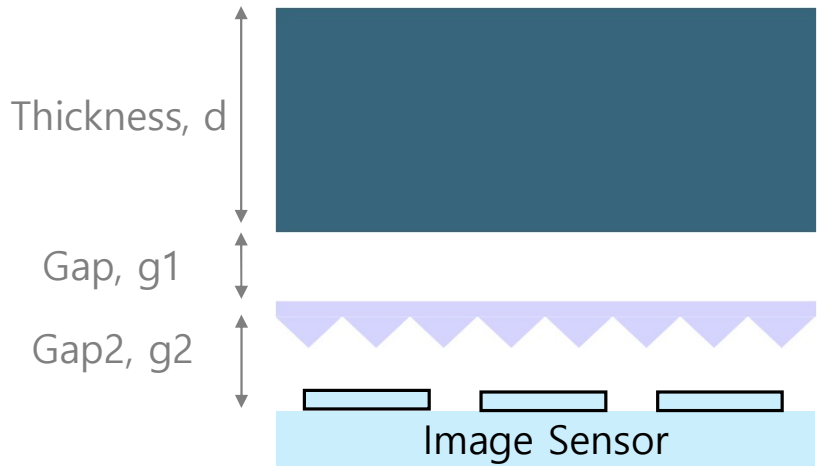
Parameter	Λ	d_x	d_y	a_{1x}, a_{1y}	a_{2x}, a_{2y}	s_{1x}	s_{2x}	s_{3x}	s_{1y}	s_{2y}	h_1	h_2
Value [nm]	545	λ	334	150	80	48.5	147	118.5	92	127	544	L

Negative Refraction



Scintillator의 최대 발광 파장인 545nm에서 음굴절률을 나타내는 Metasurface를 설계함.

Selected Parameters

	Structure 1	Structure 2	Structure 3
Figure			
Pixel Pitch	100 ㎛	100 ㎛	124 ㎛
Thickness, d	100 ㎛	200 ㎛	400 ㎛
Gap1, g1	20 ㎛		
Gap2, g2	Requires Optimization		

Scintillator의 두께에 따라 Gap2를 Optimization한 후 MTF를 계산함.

Simulation Methodology

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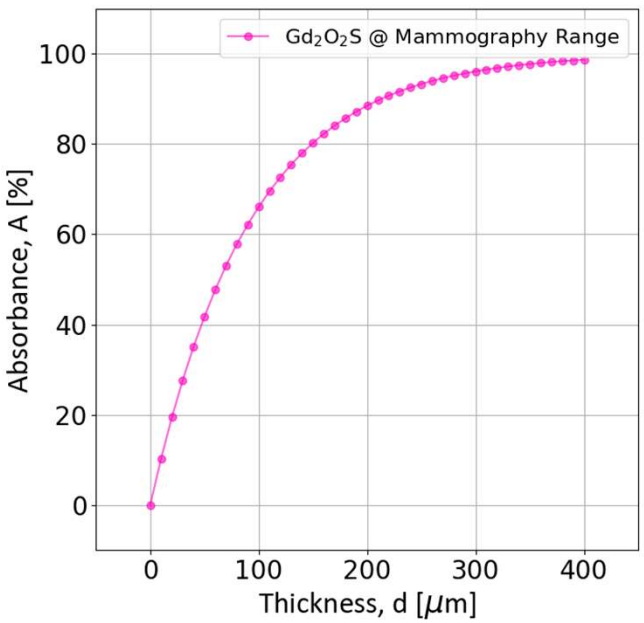
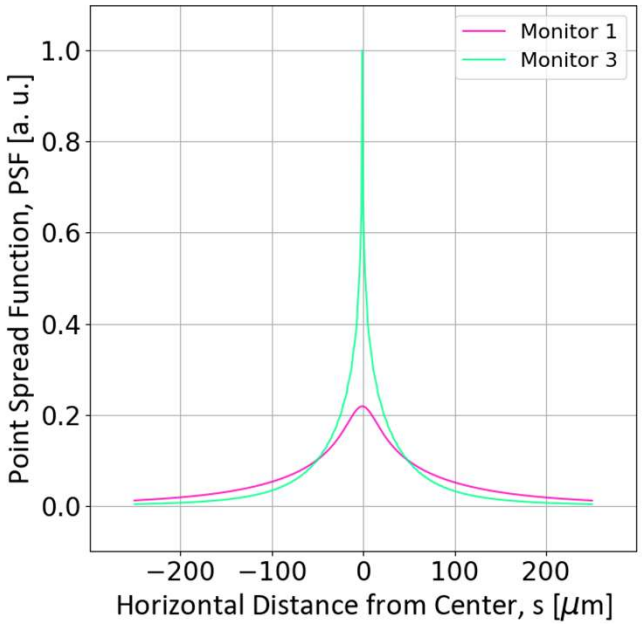
X-Ray Energy Deposition



Spherical Radiation

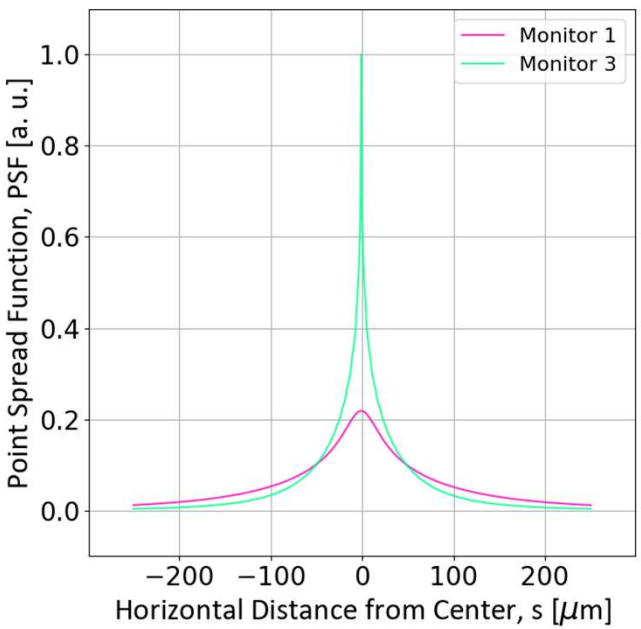
Point Spread Function



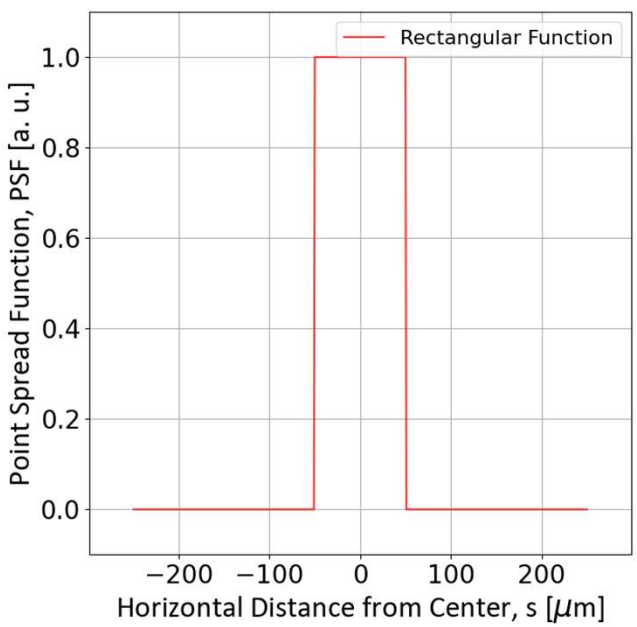
X-Ray Energy Deposition ∗ Spherical Wave Function

Simulation Methodology

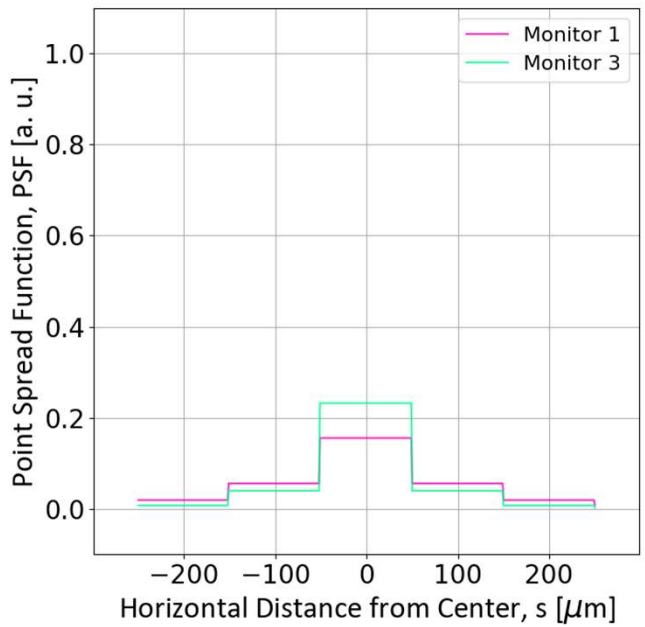
Point Spread Function



Rectangular Function



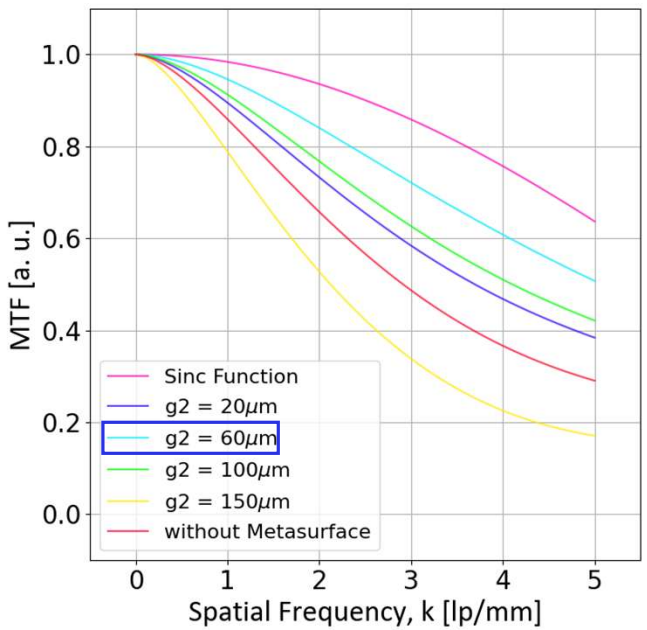
Pixelation

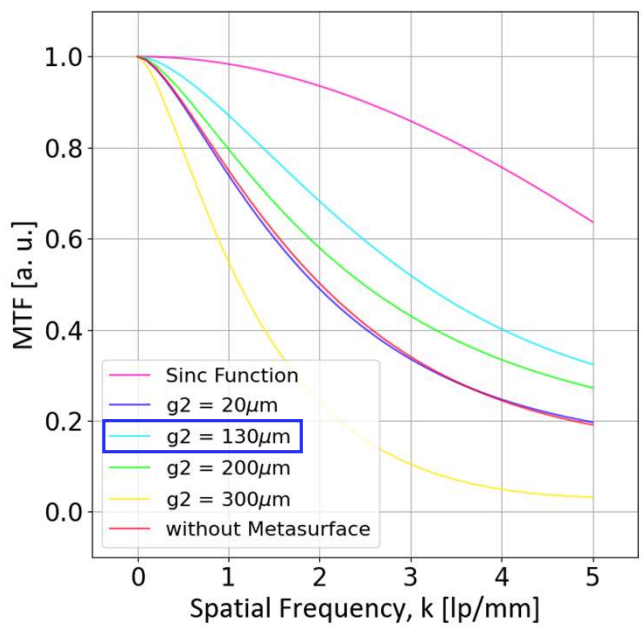


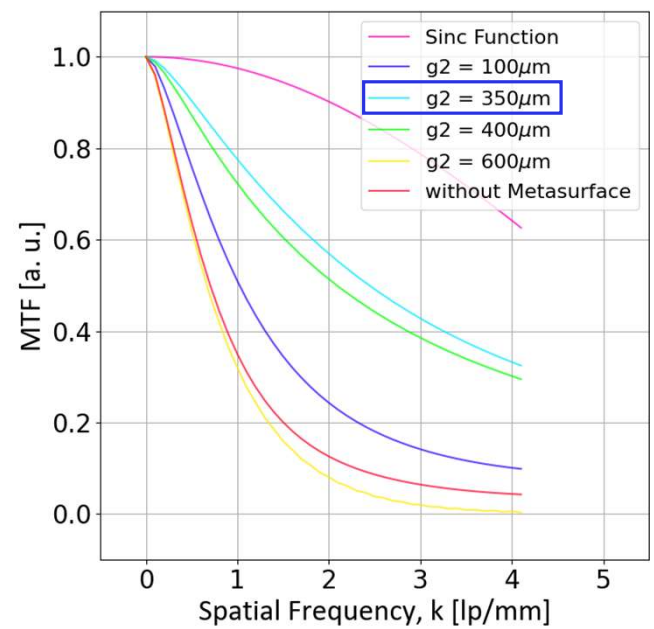
Point Spread Function × Series of Rectangular Function

Gap2 Optimization

Structure 1	Structure 2	Structure 3



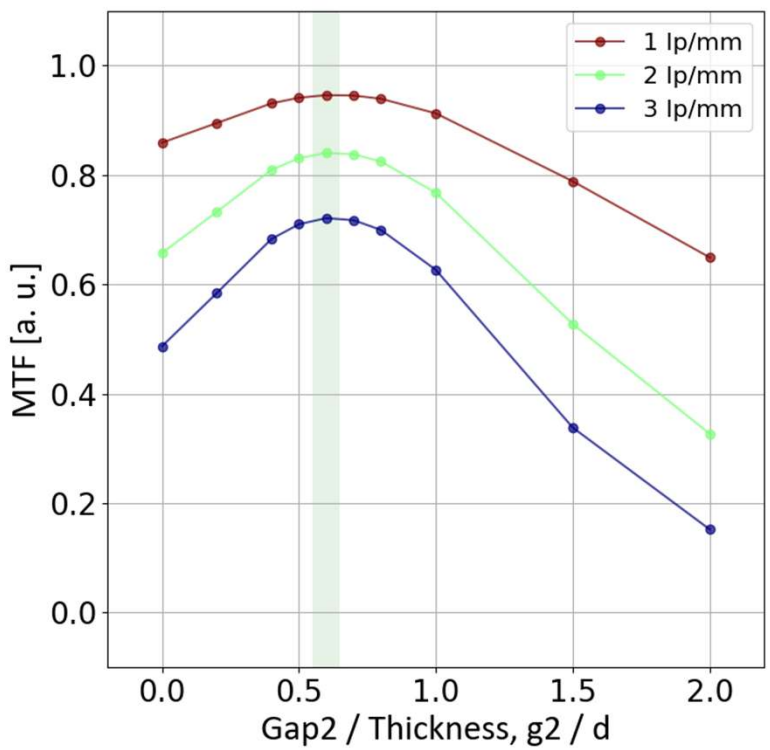




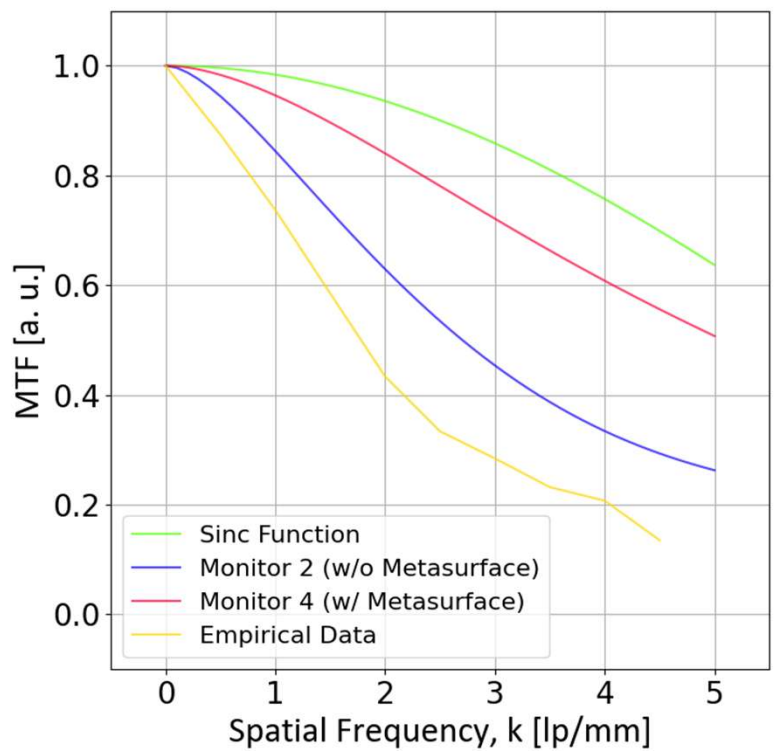
각 조건에서 MTF를 기준으로 Gap2를 Optimization ▶Gap2는 Scintillator 두께의 50~100% 수준이 적절하며, 너무 크면 역효과가 발생함.

Result: Structure 1 (Thickness = 100μm)

Gap2 Dependency

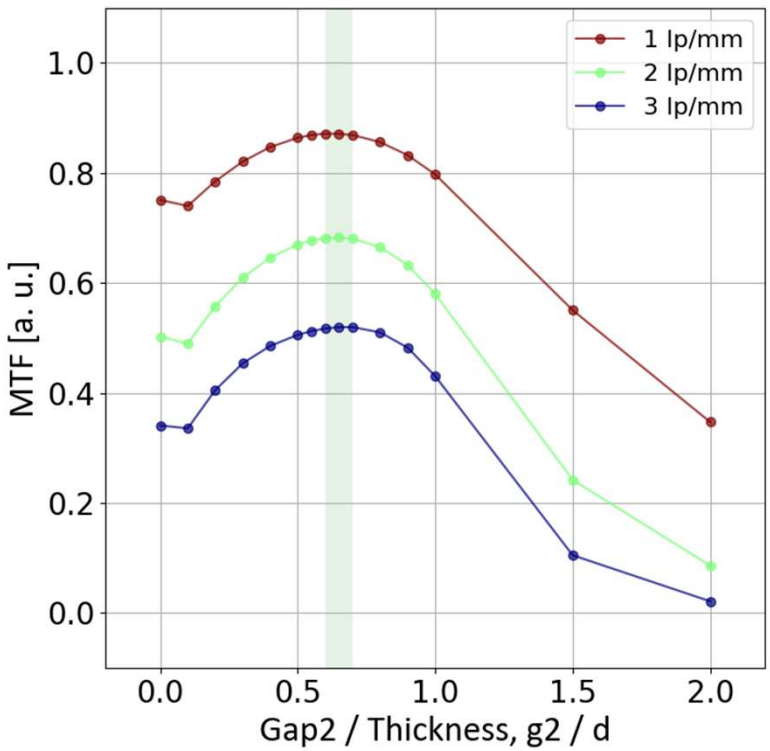


Modulation Transfer Function

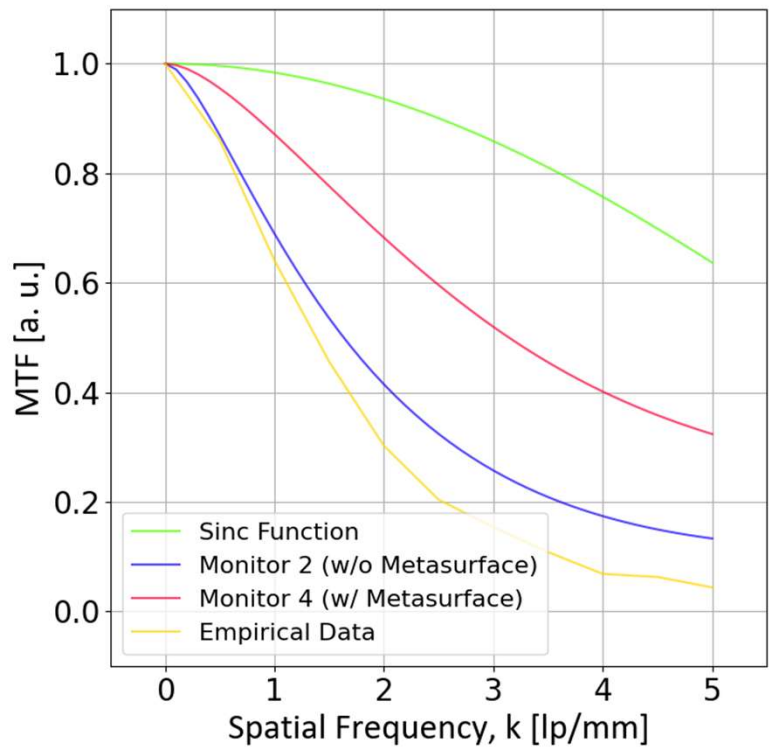


Result: Structure 2 (Thickness = 200μm)

Gap2 Dependency

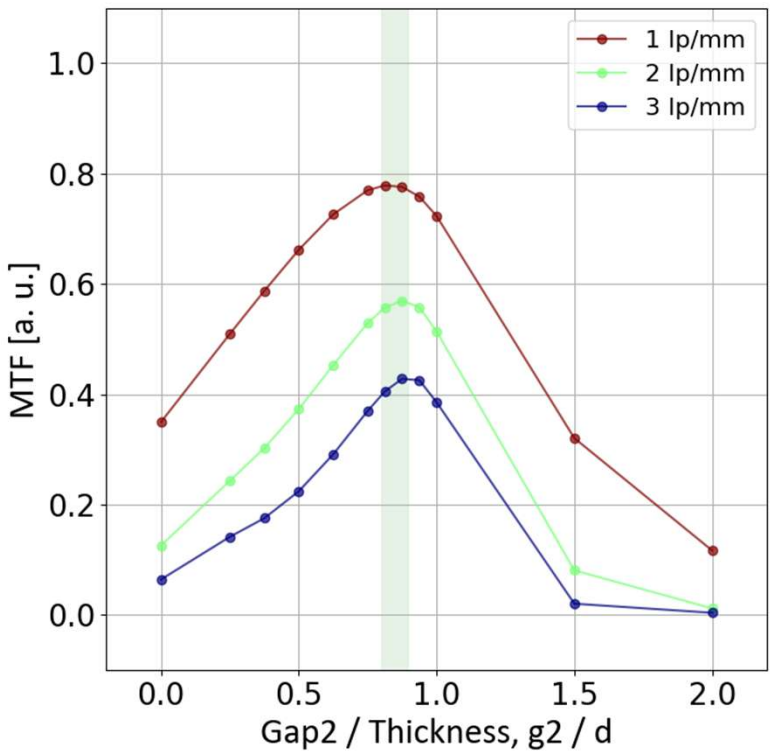


Modulation Transfer Function

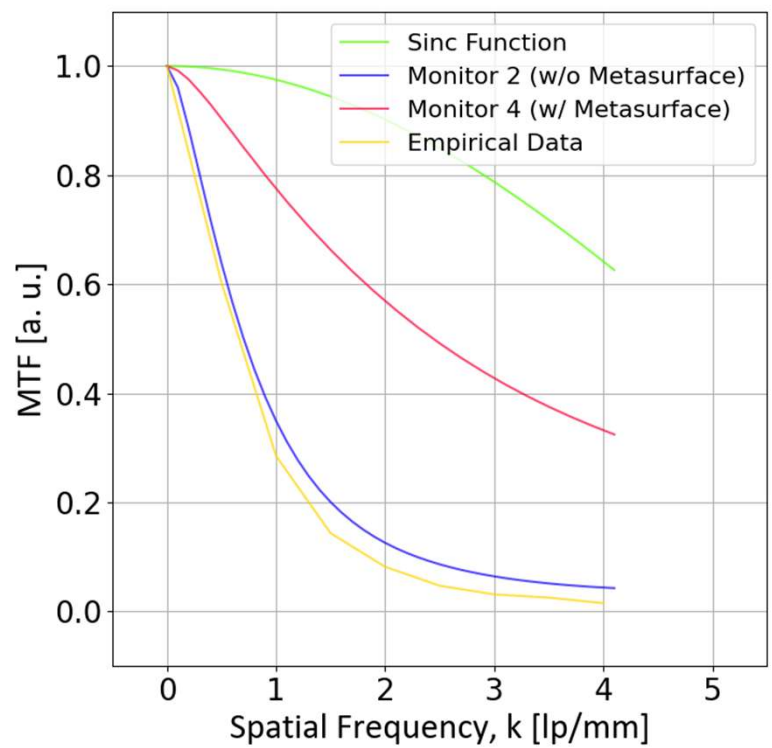


Result: Structure 3 (Thickness = 400μm)

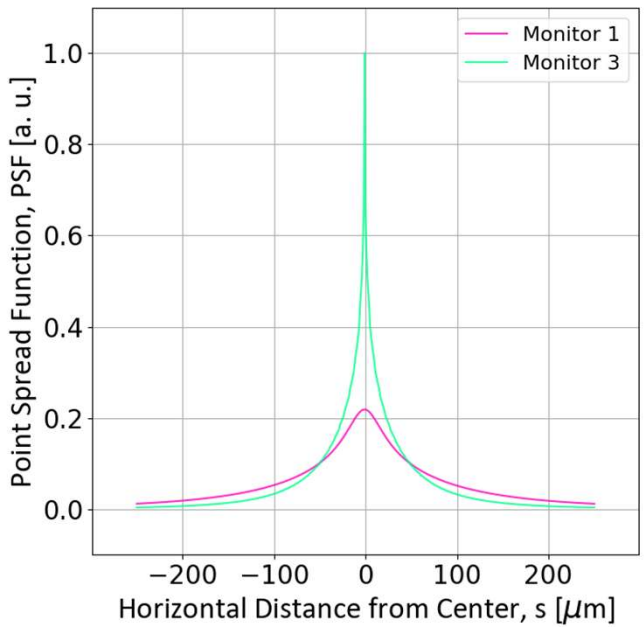
Gap2 Dependency

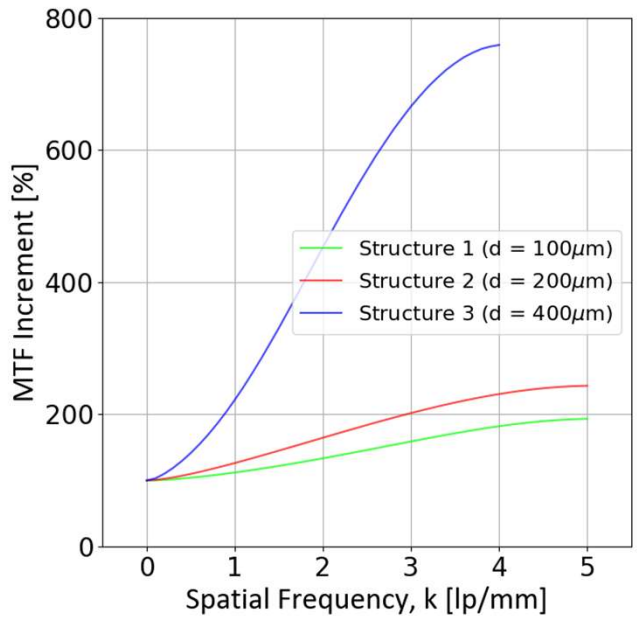


Modulation Transfer Function



Conclusion





1. Metasurface를 이용해 Indirect X-Ray Detector에 광학적 소자를 집적시킴으로써 MTF 향상을 기대할 수 있음.

2. 정밀한 Modelling 및 실제 구현을 위한 후속 연구는 필요함.

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감사합니다